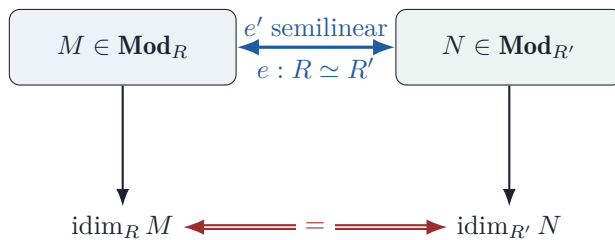


Injective dimension survives change of coordinates

A visual mathematical reading and PR review of the Lean diff



$$M \simeq_{e, \text{semi}} N \implies \mathrm{idim}_R(M) = \mathrm{idim}_{R'}(N).$$

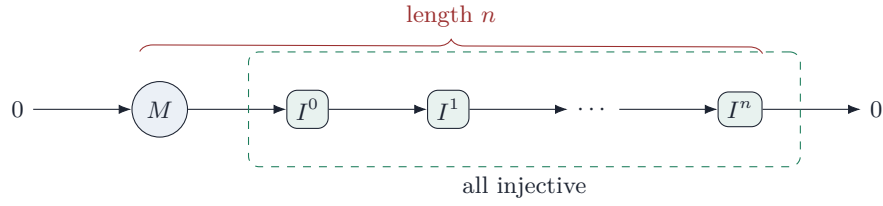
```

lemma injectiveDimension_eq_of_semiLinearEquiv [Small.{v'} R']
  {M : ModuleCat.{v} R} {N : ModuleCat.{v'} R'}
  (e' : M ≈sl[RingHomClass.toRingHom e] N) :
  injectiveDimension M = injectiveDimension N := by
  ...

```

Verdict: mathematically structural and sound in shape. **Before merge:** title fix + dead-line deletion. **Follow-up:** exact-equivalence transport API.

1. The invariant: shortest injective resolution

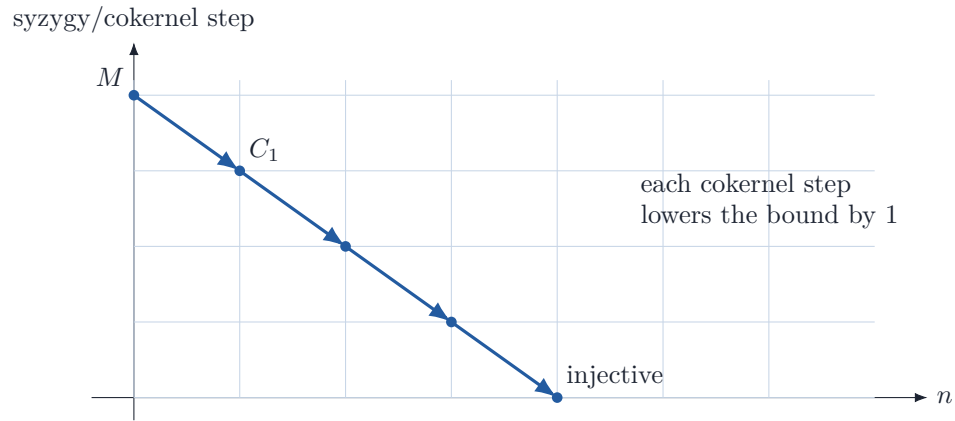


$$\text{idim}(M) \leq n \iff M \text{ admits an injective resolution of length at most } n.$$

The code uses the dimension-shifting recurrence rather than manipulating a whole resolution:

$$0 \longrightarrow M \xrightarrow{f} I \longrightarrow C \longrightarrow 0, \quad I \text{ injective}, \quad \boxed{\text{idim}(M) \leq n+1 \iff \text{idim}(C) \leq n.}$$

```
(exactS.hasInjectiveDimensionLT_X3_iff n inferInstance)
```



2. Base case: Baer transport

At $n = 0$, dimension zero means injective. A linear equivalence preserves the ideal-extension criterion (Baer's criterion):

$$\text{idim}(M) \leq 0 \iff M \text{ injective} \iff N \text{ injective} \iff \text{idim}(N) \leq 0.$$

```
simp [HasInjectiveDimensionLE, ← injective_iff_hasInjectiveDimensionLT_one,
← Module.injective_iff_injective_object R M,
← Module.injective_iff_injective_object R N,
← Module.Baer.iff_injective, Module.Baer.congr e]
```

3. Inductive engine: two short exact rows

$$\begin{array}{ccccccc}
 0 & \longrightarrow & M & \xrightarrow{f} & I & \xrightarrow{q} & C = I / \text{range}(f) \longrightarrow 0 \\
 & & \downarrow e' \sim & & \downarrow e_I \sim & & \downarrow e_C \sim \\
 0 & \longrightarrow & N & \xrightarrow{f'} & I' & \xrightarrow{q'} & C' = I' / \text{range}(f') \longrightarrow 0
 \end{array}$$

$$\boxed{\text{idim}(M) \leq n+1} \xleftrightarrow{\text{shift}} \boxed{\text{idim}(C) \leq n} \xleftrightarrow{e_C} \boxed{\text{idim}(C') \leq n} \xleftrightarrow{\text{shift}} \boxed{\text{idim}(N) \leq n+1}$$

This four-arrow chain is the proof.

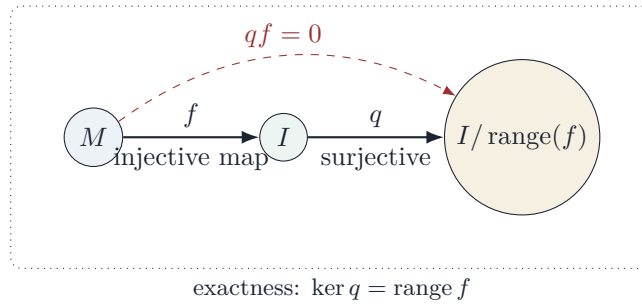
```

exact (exactS.hasInjectiveDimensionLT_X3_iff n inferInstance).symm.trans
  ((ih eCoker).trans
    (exactS'.hasInjectiveDimensionLT_X3_iff n inferInstance))
  
```

Canonical short exact sequence

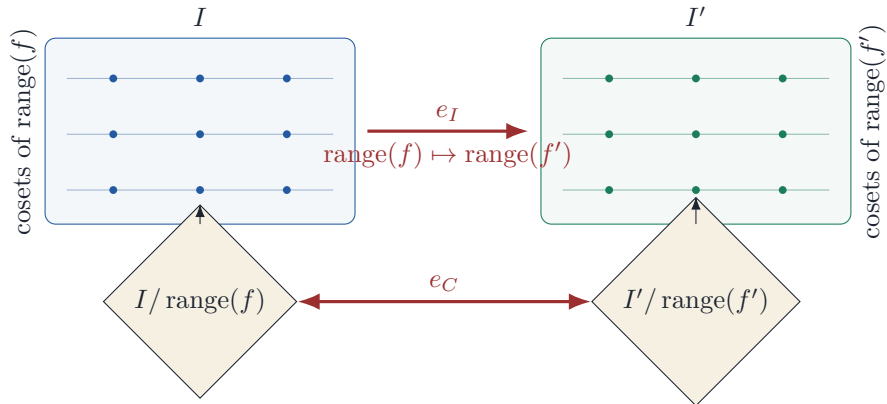
```

let S : ShortComplex (ModuleCat.{v} R) :=
  ModuleCat.shortComplexOfCompEqZero _
  f.hom.exact_map_mkQ_range.linearMap_comp_eq_zero
have exactS := ModuleCat.shortComplex_shortExact S
f.hom.exact_map_mkQ_range injf (Submodule.mkQ_surjective _)
  
```



Improvement Package this repeated construction as a canonical short-exact-sequence helper.

4. The quotient equivalence, geometrically



For classes $[i]$, the induced map is

$$e_C([i]) = [e_I(i)], \quad e_C^{-1}([i']) = [e_I^{-1}(i')].$$

Well-definedness is exactly $\text{ang}(f) \xrightarrow{e_I} \text{range}(f')$.

```

have eqmap : f'.hom.range.map el.symm.toLinearMap = f.range := by
  simp [f, LinearMap.range_comp]
let eCoker : ModuleCat.of R (I / f.range)  $\approx_{\text{sl}}$  [RingHomClass.toRingHom e]
  ModuleCat.of R' (I' / f'.hom.range) := {
    _ := Submodule.mapQ _ _ el.toLinearMap ...
    invFun := Submodule.mapQ _ _ el.symm.toLinearMap ...
    left_inv x := Submodule.Quotient.induction_on _ x (by simp)
    right_inv x := Submodule.Quotient.induction_on _ x (by simp) }

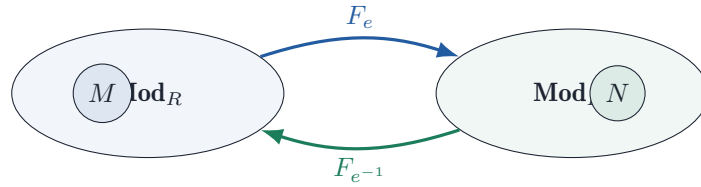
```

Improvement Add reusable API: semilinear equivalence + mapped-submodule equality $\Rightarrow$ quotient semilinear equivalence.

5. Change of scalars is a categorical equivalence

A ring equivalence changes scalar notation, not module geometry:

$$e : R \simeq R', \quad e' : M \simeq_{e, \text{semi}} N, \quad e'(rm) = e(r)e'(m).$$



additive equivalence + exactness
+ preservation of injectives

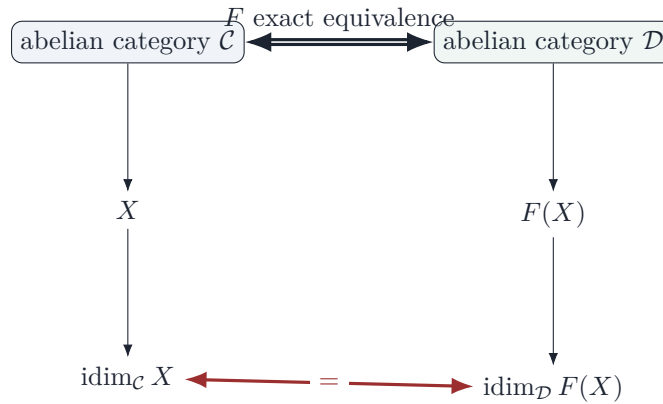
The implementation rebuilds the transported middle term:

```
let I : ModuleCat.{v} R :=
  let := Module.compHom I' e.toRingHom
  ModuleCat.of R I'
let el : I ≈si [RingHomClass.toRingHom e] I' := {
  _ := AddEquiv.refl I
  map_smul' r i := rfl }
```

Injectivity crosses the equivalence:

```
have : Injective (ModuleCat.of R I) := by
  rw [← Module.injective_iff_injective_object]
  exact Module.Injective.of_ringEquiv e.symm el.symm
```

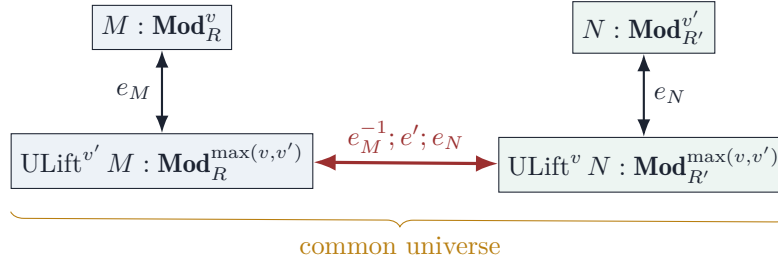
The unusually structural general theorem



$$F : \mathcal{C} \simeq \mathcal{D} \text{ exact} \implies \text{idim}_{\mathcal{C}}(X) = \text{idim}_{\mathcal{D}}(FX).$$

Design direction Prove this once in the abelian-category API; derive the module theorem. It removes manual transport of injectives, ranges, and quotient representatives.

6. Universe bridge: move both objects to one floor



```

let eM : M ≈[R] ModuleCat.of R (ULift.{v'} M) := ULift.moduleEquiv.symm
let eN : N ≈[R'] ModuleCat.of R' (ULift.{v} N) := ULift.moduleEquiv.symm
rw [hasInjectiveDimensionLE_iff_of_linearEquiv_aux eM,
    hasInjectiveDimensionLE_iff_of_linearEquiv_aux eN]
...
exact hasInjectiveDimensionLE_iff_of_semiLinearEquiv_aux e
      ((eM.symm.trans e').trans eN) n

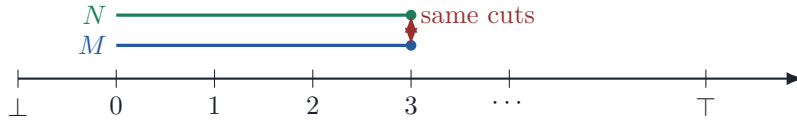
```

Keep This is explicit, readable, and preferable to opaque coercion tricks.

7. From all bounds to equality in $\mathbb{N}_\infty$

The dimension takes values in extended naturals. Equality is tested by all upper cuts:

$$a = b \iff \forall d \in \mathbb{N}_\infty, (d \leq a \iff d \leq b).$$



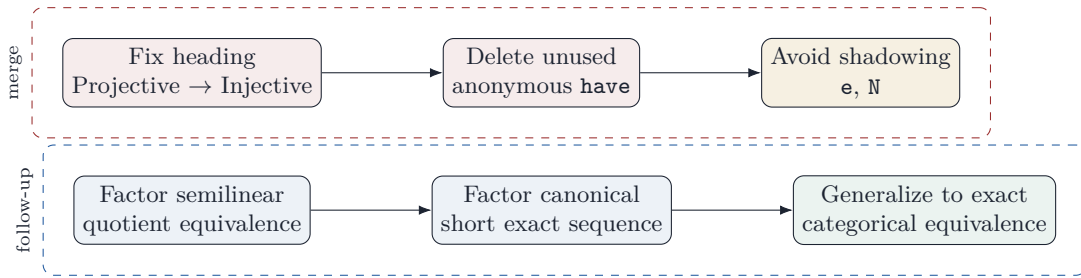
```

refine eq_of_forall_ge_iff (fun N => ?_)
  induction N with
  | bot => simp [injectiveDimension_eq_bot_iff,
    ModuleCat.isZero_iff_subsingleton] using e'.subsingleton_congr
  | coe n =>
    induction n with
    | top => simp
    | coe n =>
      norm_cast
      simp only [injectiveDimension_le_iff]
      exact hasInjectiveDimensionLE_iff_of_semiLinearEquiv e e' n

```

Naming Rename the cutoff N to d ; N already names a module.

8. Audit map



Delete

```

have := exactS.hasInjectiveDimensionLT_X3_iff
n inferInstance

```

It is immediately repeated and unused.

Preserve

```

lemma injectiveDimension_eq_of_linearEquiv ... :=
  injectiveDimension_eq_of_semiLinearEquiv
  ... (RingEquiv.refl R) e

```

Excellent API layering: no duplicated proof.

9. Mathematical proof, compressed

Theorem. Let $e : R \simeq R'$ be an equivalence of commutative rings and $\phi : M \simeq_{e, \text{semi}} N$. Then $\text{idim}_R M = \text{idim}_{R'} N$.

Proof. For every $n \in \mathbb{N}$, prove $\text{idim}_R M \leq n \iff \text{idim}_{R'} N \leq n$ by induction. For $n = 0$, use preservation of injectivity under semilinear equivalence. For $n + 1$, choose $0 \rightarrow N \rightarrow I' \rightarrow C' \rightarrow 0$ with I' injective. Transport it through e^{-1} and ϕ to $0 \rightarrow M \rightarrow I \rightarrow C \rightarrow 0$; then I is injective and $C \simeq C'$ semilinearly. Dimension shifting and induction give

$$\text{idim}_R M \leq n + 1 \iff \text{idim}_R C \leq n \iff \text{idim}_{R'} C' \leq n \iff \text{idim}_{R'} N \leq n + 1.$$

If the module universes differ, replace M, N by their linearly equivalent ULifts. Finally, equality in $\mathbb{N}_\infty$ follows from equality of every finite upper bound; the bottom case is preserved because ϕ preserves subsingletons. $\square$

